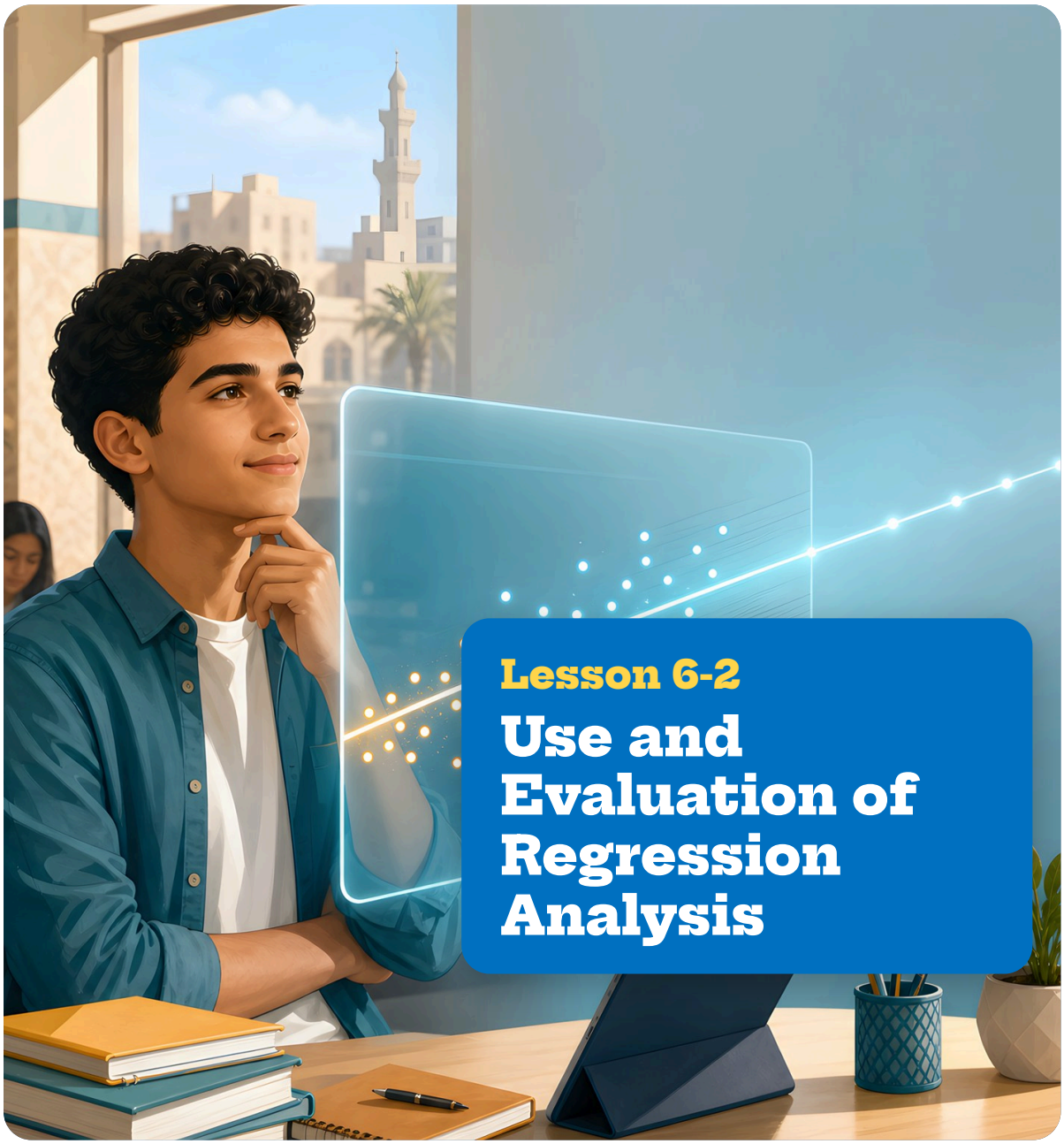


Learning Objectives

- Explain the difference between simple and multiple regression, and what a regression model represents.
- Evaluate a regression model using R^2, RMSE, and MAE, and explain why a good fit does not prove causation.

Data often hides a relationship you can turn into a prediction: the warmer the day, the more ice cream sells. Regression analysis writes that relationship as an equation, so you can put in tomorrow's forecast temperature and predict tomorrow's sales. Yet a prediction equation is only a model — a simplified picture — and a careful analyst asks two more questions: how accurate is it (using R^2, RMSE, and MAE), and does "fits well" really mean "causes"? A line that fits the data is not proof of cause.

? Guiding question: How does regression turn data into a prediction, and how do we judge whether that prediction can be trusted?

Point!

1. The Regression Analysis Stage

(1) **(Regression analysis)**…a method that expresses the relationship between variables in the data as an equation, and uses it for prediction and simulation. For example, from past data on temperature and sales, one can express how sales change when temperature rises as an equation, and put tomorrow's forecast temperature into that equation to predict sales.

(2) In regression analysis, the variable that is the result is called the **(response variable) (dependent variable)**, and the variable used to explain or predict it is called the **(explanatory variable)**. In the temperature-and-sales example, sales are the response variable and temperature is the explanatory variable.

(3) The equation obtained from data in regression analysis is called the **(model)**. A model is a simplified representation of a real relationship—for example, putting the trend "the higher the temperature, the higher the sales" into a format that can be calculated.

(4) Both creating a model and evaluating whether the model can actually be used are important. A model that is only created is not useful; only by confirming its accuracy and limitations does it lead to practical use.

2. Simple and Multiple Regression

(1) **(Simple regression analysis)**…a method of predicting the response variable using one explanatory variable.

e.g., Predict ice cream sales (y) from temperature (x) alone. The prediction equation is expressed as a straight line in the format **$y = ax + b$**, and this line is called the **(regression line)**. a and b are determined by the **(least-squares method)** (a method that determines the line so as to minimize the sum of the squares of the discrepancies between the actual values and the regression line, each called a **(residual)**).

(2) **(Multiple regression analysis)**…a method of predicting the response variable using multiple explanatory variables. In real data, results are rarely determined by a single cause, and multiple regression analysis, which can consider several factors simultaneously, is suited to more realistic prediction.

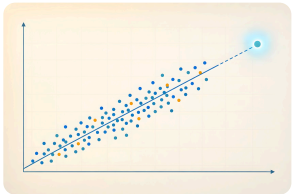
Key Fact:
Regression analysis expresses the relationship between an explanatory variable (the input used to make the prediction) and a response variable (the result) as a model — a line $y = ax + b$ for simple regression, or a sum of weighted variables for multiple regression. Judge the model with R^2, RMSE, and MAE, and never mistake a good fit for proof of causation.

Key Concepts:

- regression analysis
- response / explanatory variable
- simple / multiple regression
- regression line, least-squares
- R^2, RMSE, MAE
- correlation versus causation

KEY TERMS

regression analysis — expressing a relationship as an equation for prediction.
response variable — the result (dependent).
explanatory variable — the input used to make the prediction.



Fit a line to data, then use it to predict within that range.

e.g., Predict housing prices from a house's size, age, and distance from a transit station (multiple factors). The prediction equation is expressed as the sum of multiple explanatory variables, each multiplied by a coefficient (a numerical value representing the strength of its influence), plus an intercept. For example, when there are two explanatory variables, the format becomes $\boldsymbol{y = a_1x_1 + a_2x_2 + b}$ (x_1 and x_2 are the explanatory variables, a_1 and a_2 are their coefficients, and b is the intercept). When there are three or more explanatory variables, terms are added in the same way.

(3) The following differences exist between simple and multiple regression.

Aspect	Simple regression	Multiple regression
Number of explanatory variables	One	Two or more
Characteristics of the model	Simple and easy to interpret	Can express complex relationships
Suitable situations	When you want to capture the relationship between two variables concisely	When the response variable is influenced by multiple factors

3. Evaluation Indicators for Models

(1) To judge how useful a model created by regression analysis is, **(evaluation indicators)** are used. Three typical indicators are as follows.

Indicator	Description (concept of calculation)	Interpretation
(R^2 (coefficient of determination))	Expresses, on a 0–1 scale, how well the model explains the variability in the data	The closer to 1, the better the variation in the data is explained
(RMSE (root mean square error))	The square root of the mean of the squared discrepancies between predicted and actual values	The smaller the value, the higher the prediction accuracy; expressed in the same units as the response variable
(MAE (mean absolute error))	The mean of the absolute values of the discrepancies between predicted and actual values	The smaller the value, the higher the prediction accuracy; less affected by extremely large discrepancies

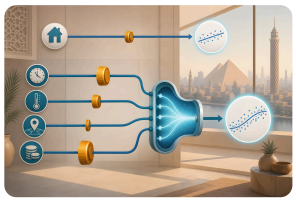
(2) Both RMSE and MAE indicate the size of prediction errors, but RMSE tends to weight large discrepancies more heavily. On the other hand, MAE evaluates discrepancies on average, so it is less affected by outliers. They are used differently according to the purpose of the analysis.

4. Cautions When Using Regression Models

(1) Regression analysis is a widely applicable method, but misuse can lead to incorrect conclusions. Pay particular attention to the following points.

PAUSE & THINK

One cause or several? Which kind of regression would house prices need?



Multiple regression combines several inputs.

KEY TERMS

R^2 — how well the model explains the variability in the data (0–1).
RMSE — error size, weights big misses more.
MAE — average error, less swayed by outliers.

PAUSE & THINK

A prediction line fits the data well — does that prove one variable causes the other?

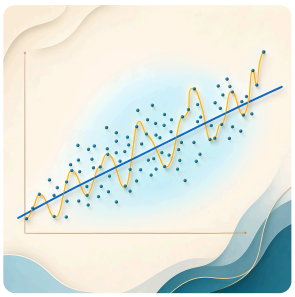
① Do not confuse correlation with causation: a prediction equation that fits does not establish it. Causation needs temporal precedence, exclusion of other factors' influence, and theoretical support.

② A model is not finished once created: Confirm accuracy with evaluation indicators, verify validity in light of the purpose, and do not apply it outside the range of the data it was fitted to.

③ Do not judge a model by R^2 alone: Make judgments together with other indicators.

④ Adding more explanatory variables does not necessarily improve a model: Appropriate variables must be chosen according to the purpose.

(2) The results of regression analysis should not be treated as absolute truth as is, but as one perspective that the data offers. Interpreting results appropriately and using them in real-world judgment is important.



An overfit model chases noise, not the trend.

PAUSE & THINK

Which error measure would you trust more when a few huge outliers exist?

Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Multiple regression analysis is a method of predicting the response variable using one explanatory variable.

B R^2 (coefficient of determination) is an indicator showing how well the model explains the variability in the data.

C The equation obtained from data in regression analysis is called the model, and is a simplified representation of a real relationship.

D If the prediction equation fits, it can be said that there is a causal relationship between the explanatory variable and the response variable.

(2) Match each description (a–c) with the most appropriate indicator from the choices below (A–C).

a The mean of the absolute values of the discrepancies between predicted and actual values

b The square root of the mean of the squared discrepancies between predicted and actual values

c How well the model explains the variability in the data

[Choices] **A** RMSE **B** R^2 **C** MAE

Solution

(1)

A Predicting with one explanatory variable is simple regression analysis. Multiple regression analysis uses multiple explanatory variables. Therefore, ×.

B Correct description of R^2. Therefore, ○.

C Correct definition of a model. Therefore, ○.

D Even if the prediction equation fits, that alone does not establish causation. Therefore, ×.

(2)

a: The mean of absolute discrepancies is MAE. **C**

b: The square root of the mean of squared discrepancies is RMSE. **A**

c: An indicator that explains the variability in the data is R^2. **B**

Try

1 Answer the following questions.

1. What is the term for the regression analysis that predicts the response variable using multiple explanatory variables?
2. What is the term for the indicator showing how well the model explains the variability in the data?
3. In regression analysis, what are the variables called that are the result and the input used to predict it, respectively?
4. What is the term for "the discrepancy between actual values and the regression line" used in simple regression analysis?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the difference between simple and multiple regression.

A Simple regression can express complex relationships, while multiple regression is simple.

B Simple regression uses multiple explanatory variables, and multiple regression uses one explanatory variable.

C Simple regression uses one explanatory variable, and multiple regression uses multiple explanatory variables.

D There is no difference between simple and multiple regression in the number of explanatory variables; only the calculation method differs.

(2) For each of analysis purposes (a, b), choose the most appropriate regression analysis method from the choices below (A, B).

a You want to predict ice cream sales from temperature (one variable).

b You want to predict housing prices from a house's size, age, and distance from a transit station (multiple variables).

[Choices] **A** Simple regression analysis **B** Multiple regression analysis

EXAM-STYLE QUESTION

Explain why a regression model with a high R^2 can still be a poor basis for claiming that one variable causes another. **[6] Refer to:** correlation versus causation.

PAUSE & THINK

One variable in, or several? That is the whole difference.

(3) From the following options (A–D), choose the one that most appropriately describes the interpretation of regression analysis results.

A If a prediction equation is obtained from regression analysis, the equation necessarily indicates a causal relationship.

B The results of regression analysis should be treated as one perspective that the data offers, and whether causation has been demonstrated must be carefully evaluated.

C If R^2 is high, the model is necessarily a good model.

D If the prediction equation fits, one may conclude that the explanatory variable is the cause of the response variable.

(4) From the following options (A–D), choose the one that is NOT an appropriate description of evaluation indicators for regression models.

A R^2 (coefficient of determination) takes values between 0 and 1; the closer to 1, the better the variation in the data is explained.

B RMSE is the square root of the mean of squared discrepancies between predicted and actual values, expressed in the same units as the response variable.

C MAE is the mean of the absolute values of the discrepancies between predicted and actual values, and is less affected by extremely large discrepancies.

D If the value of R^2 is close to 1, the model is necessarily a good model without needing to check other indicators.

PAUSE & THINK

A high R^2 alone — enough to trust a model, or do you need RMSE and MAE too?

Think as an Engineer — Investigate & Decide

Your school canteen wants to predict its daily cold-drink sales so it can order the right amount. You plan a regression model.

☐ **Trust one obvious cause** ☐ **Model it and test the model**

1 · Set up the model. Name the response variable, then list the explanatory variables you would try (for example the day's temperature, the number of students present, and whether it is an exam day). Decide whether this is a simple or a multiple regression, and say why.

2 · Judge and question it. Choose which evaluation indicator you would report (R^2, RMSE, or MAE) and justify it for this purpose. Then name one confounder (a confounding variable) — a hidden third factor that could make two variables move together without one causing the other — that could mislead you here.

3 · Decide and interpret. Suppose your model gives a high R^2. State whether that alone would let you claim "hot days cause higher sales," and what else you would need. **In pairs,** compare models and agree on the single most useful explanatory variable to keep.

Give evidence for your answer.

KEY TERMS

coefficient — the weight on an explanatory variable.
intercept — the constant *b* in the equation.
confounder — a hidden third cause behind a correlation.

Stuck? Start from the result you want to predict (sales), then ask which measurable things plausibly move it — and for each, ask "could something else be driving both?" A high R^2 tells you the line fits, not that the cause is real.

🌍 In a New Context

A city finds that months with more ice cream sales also have more swimming accidents, and a regression line fits the two closely. Explain why this does not mean ice cream causes accidents, and name the hidden factor most likely behind both.

❓ Lesson Question — Answered

The lesson asked how regression turns data into a prediction and how we judge whether it can be trusted. Regression analysis expresses the relationship between an explanatory variable (the input used to make the prediction) and a response variable (the result) as a model: a regression line $y = ax + b$ in simple regression (one explanatory variable), or a weighted sum such as $y = a_1x_1 + a_2x_2 + b$ in multiple regression (several variables), fitted by least squares to minimize the sum of squared residuals. Trust is judged with evaluation indicators used together — R^2 for how much variability is explained, and RMSE and MAE for the size of prediction errors (RMSE weighting large misses more, MAE less swayed by outliers) — never R^2 alone, and never by adding variables for their own sake. Above all, a good fit is only a correlation: causation requires temporal precedence, ruling out other factors, and theory. Regression offers one perspective from the data, to be interpreted, not obeyed.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

In regression analysis, the variable that is the result is called the (a), and the variable used to predict it is called the (b). The method that predicts using one

explanatory variable is called (c), and the method that predicts using multiple explanatory variables is called (d). To judge whether a model is good, evaluation indicators are used. Indicators include (e), which shows how well the model explains the variability in the data, and (f) and (g), which show the size of the discrepancies between predicted and actual values. To claim causation from the results of regression analysis, temporal precedence, exclusion of other factors' influence, theoretical support, and so on are necessary, and one must not conclude that there is (h) just because the prediction equation fits.

1. Fill in the blanks (a)–(h).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes correlation and causation.

A If there is correlation, causation necessarily holds.

B To claim causation, temporal precedence, exclusion of other factors' influence, theoretical support, and so on are necessary.

C If a prediction equation is obtained from regression analysis, the equation completely demonstrates causation.

D If there is no correlation, it can be said that there is no relationship at all between the two variables.

(2) From the following options (A–D), choose the one that most appropriately describes the interpretation of evaluation indicators.

A R^2 takes values between 0 and 1; the closer to 1, the better the variation in the data is explained.

B RMSE and MAE use the same calculation method, and their values always coincide.

C A model with a high R^2 is always a good model in any situation.

D MAE is an indicator that weights large discrepancies more heavily.

(3) From the following options (A–D), choose the one that is NOT an appropriate point of caution when using regression models.

A Do not conclude that there is causation just because the prediction equation fits.

B Confirm accuracy with evaluation indicators and verify the validity of the model in light of the purpose.

C If R^2 is close to 1, one may judge the goodness of the model without checking other indicators.

D Adding more explanatory variables does not necessarily improve accuracy. Appropriate variables must be chosen according to the purpose.

REFLECT & CHALLENGE

Reflect: Think of two things you have seen rise together (say, screen time and poor sleep). What hidden factor could drive both? **Challenge:** You may add only one more explanatory variable to a model that already has a high R^2. Argue whether you would add it at all, and state the one check you would run before trusting the bigger model.

⭐ Key takeaway

Remember:
Regression predicts a response variable from one (simple, $y = ax + b$) or several (multiple) explanatory variables, fitted by least-squares. Judge it with R^2, RMSE, and MAE together — and never read a good fit as proof of cause.